PS2

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2026-03-03

GitHub Repository URL: https://github.com/jennyjia319/gov-51-ps2

1 Part 1: The Original Study

1.1 Exploring the Experiment

1.1.1 Load Packages and Data

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.2     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
gay <- read.csv("data/raw/gay.csv")
```

Q1.1

```
nrow(gay) # look at total observations
```

```
## [1] 69592
```

```
n_distinct(gay$study) # look at unique values
```

```
## [1] 2
```

```r
unique(gay$treatment) #list treatment conditions
```

```
## [1] "No Contact"
## [2] "Recycling Script by Gay Canvasser"
## [3] "Same-Sex Marriage Script by Gay Canvasser"
## [4] "Recycling Script by Straight Canvasser"
## [5] "Same-Sex Marriage Script by Straight Canvasser"
```

There are 69,592 total observations in the dataset. There are 2 unique values of study. The treatment conditions are: "no contact", "recycling script by gay canvasser", "same-sex marriage script by gay canvasser", "recycling script by straight canvasser", and "same-sex marriage script by straight canvasser".

Q1.2

```r
study1wave1 <- gay |> # create new dataset that only includes study 1 and wave 1
  filter(study == 1, wave == 1)
mean <- study1wave1 |>
  group_by(treatment) |>
  summarize(mean_ssm = mean(ssm, na.rm = TRUE),
  n = n() # counting observations in each treatment group
) |>
  arrange(desc(mean_ssm)) # arrange for easier viewing
mean # to show the highest means
```

```
## # A tibble: 5 x 3
##   treatment                                        mean_ssm     n
##   <chr>                                               <dbl> <int>
## 1 Recycling Script by Gay Canvasser                    3.13  1046
## 2 Same-Sex Marriage Script by Straight Canvasser       3.10  1033
## 3 No Contact                                           3.04  5238
## 4 Same-Sex Marriage Script by Gay Canvasser            3.03  1151
## 5 Recycling Script by Straight Canvasser               3.01  1039
```

Within study 1 wave 1, treatment group "Recycling Script by Gay Canvasser" shows the highest average support, with a mean of 3.130975, and 1046 observations. The other means are: 3.099710 for Same-Sex Marriage by Straight Canvasser with 1033 observations, 3.042764 for No Contact with 5238 observaions, 3.025195 for Same-Sex Marriage Script by Gay Canvasser with 1151 observations, and 3.013474 for Recycling Script by Straight Canvasser with 1039 observations.

Q1.3

```r
gay |>
  filter(study == 1, wave == 1) |>
  group_by(treatment) |> #separate by treatment group
  summarize(sd_ssm = sd(ssm, na.rm = TRUE)) # SD calculation
```

```
## # A tibble: 5 x 2
##   treatment                                        sd_ssm
```

```
##   <chr>                                          <dbl>
## 1 No Contact                                      1.68
## 2 Recycling Script by Gay Canvasser               1.70
## 3 Recycling Script by Straight Canvasser          1.68
## 4 Same-Sex Marriage Script by Gay Canvasser       1.67
## 5 Same-Sex Marriage Script by Straight Canvasser  1.68
```

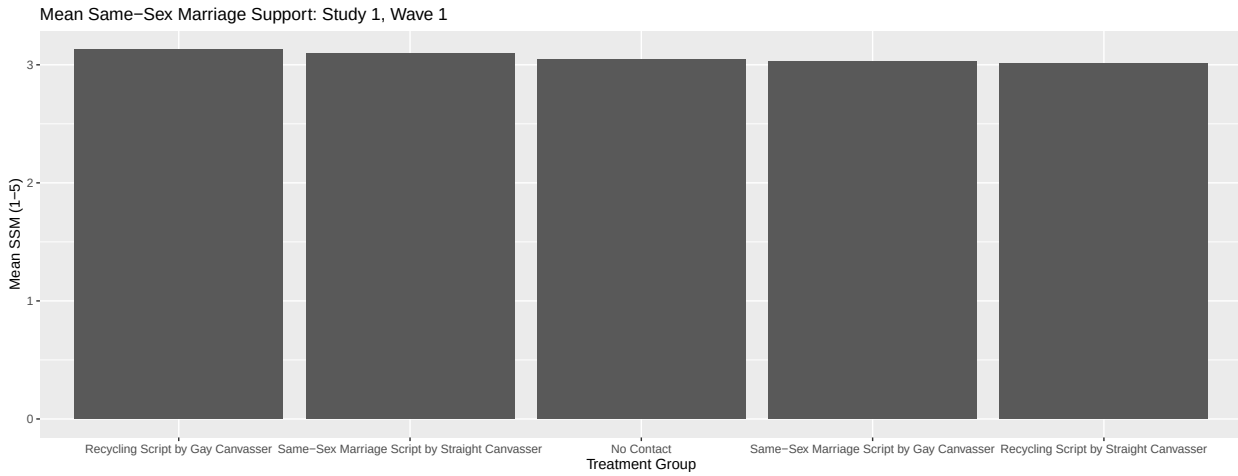
The standard deviations are: 1.680540 for No Contact 1.702533 for Recycling Script by Gay Canvasser 1.683483 for Recycling Script by Straight Canvasser 1.666099 for Same-Sex Marriage Script by Gay Canvasser 1.682885 for Same-Sex Marriage Script by Straight Canvasser. These standard deviations fall within the narrow range of 1.67-1.70. What this indicates is that the variability in attitudes is comparable across the 5 groups. This is positive, and reflects the assumption that randomization creates balanced groups. In the case that a group had a noticeably larger or smaller standard deviation, it would complicate how we can determine the fact that the treatment is solely responsible for the differences in means.

Q1.4

```r
study1wave1_means <- gay |>
  filter(study == 1, wave == 1) |>
  group_by(treatment) |>
  summarize(mean_ssm = mean(ssm, na.rm = TRUE)
  )
study1wave1_means
```

```
## # A tibble: 5 x 2
##   treatment                                     mean_ssm
##   <chr>                                            <dbl>
## 1 No Contact                                        3.04
## 2 Recycling Script by Gay Canvasser                 3.13
## 3 Recycling Script by Straight Canvasser            3.01
## 4 Same-Sex Marriage Script by Gay Canvasser         3.03
## 5 Same-Sex Marriage Script by Straight Canvasser    3.10
```

```r
#changed dimensions because default dimensions were too narrow
ggplot(study1wave1_means,
       aes(x = reorder(treatment, -mean_ssm), y = mean_ssm)) +
  geom_col() +
  labs(
    title = "Mean Same-Sex Marriage Support: Study 1, Wave 1",
    x = "Treatment Group",
    y = "Mean SSM (1-5)"
  )
```



In general, the means are so similar that it's hard to see the differences on the graph.

1.2 Checking Balance and Comparing Outcomes

Q1.5

```
study1wave1_check <- gay |>
  filter(study == 1, wave == 1,
    treatment %in% c(
      "Same-Sex Marriage Script by Gay Canvasser",
      "No Contact")
  )
stats <- study1wave1_check |>
  group_by(treatment) |>
  summarize(
    mean_ssm = mean(ssm, na.rm = TRUE),
    sd_ssm = sd(ssm, na.rm = TRUE)
  )

stats
```

```
## # A tibble: 2 x 3
##   treatment                                   mean_ssm sd_ssm
##   <chr>                                          <dbl>  <dbl>
## 1 No Contact                                      3.04   1.68
## 2 Same-Sex Marriage Script by Gay Canvasser       3.03   1.67
```

```
mean_gayscript <- stats$mean_ssm[
  stats$treatment == "Same-Sex Marriage Script by Gay Canvasser"
]

mean_nocontact <- stats$mean_ssm[
  stats$treatment == "No Contact"
]

sd_gayscript <- stats$sd_ssm[
```

```r
  stats$treatment == "Same-Sex Marriage Script by Gay Canvasser"
]

sd_nocontact <- stats$sd_ssm[
  stats$treatment == "No Contact"
]

SDM <- (mean_gayscript - mean_nocontact) /
  sqrt((sd_gayscript^2 + sd_nocontact^2) / 2)

SDM
```

```
## [1] -0.01049935
```

THe means are 3.042764 for the No Contact group and 3.025195 for the Same-Sex Marriage Script by Gay Canvasser group. The SDs are 1.680540 for the No Contact group and 1.666099 for the Same-Sex Marriage Script by Gay Canvasser Group. A standardized difference of -0.01049935 is achieved. This is very close to 0 and falls well below the 0.25 threshold for balance. What this means is that, at a baseline, Same-sex Marriage Script by Gay Canvasser group and the No Contact group are nearly identical in terms od attitudes. Thus, randomization created balanced groups.

Q1.6

```r
study1wave2_check <- gay |>
  filter(study == 1, wave == 2,
    treatment %in% c(
      "Same-Sex Marriage Script by Gay Canvasser",
      "No Contact")
  )
wave2_stats <- study1wave2_check |>
  group_by(treatment) |>
  summarize(
    mean_ssm = mean(ssm, na.rm = TRUE)
  )
wave2_stats
```

```
## # A tibble: 2 x 2
##   treatment                                  mean_ssm
##   <chr>                                         <dbl>
## 1 No Contact                                     3.04
## 2 Same-Sex Marriage Script by Gay Canvasser      3.14
```

```r
mean_gayscript_w2 <- wave2_stats$mean_ssm[
  wave2_stats$treatment == "Same-Sex Marriage Script by Gay Canvasser"]

mean_nocontact_w2 <- wave2_stats$mean_ssm[
  wave2_stats$treatment == "No Contact"]

SDM2 <- mean_gayscript_w2 - mean_nocontact_w2

SDM2
```

```
## [1] 0.09987463
```

Now looking at Study 1 Wave 2, the mean ssm for No Contact is 3.039615 and 3.139489 for Same-sex Marriage Script by Gay Canvasser. The difference in means is 0.09987463 points. The difference at wave 2 must suggest, then, that contact with a gay canvasser script moderately shifted attitudes on same-sex marriage relative to the no contact group. # 2 Part 2: The Forensic Discovery ## 2.1 The Baseline Distribution ### Q2.1

```r
gay_r <- read_csv("data/raw/gayreshaped.csv")
```

```
## Rows: 11948 Columns: 6
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): treatment
## dbl (5): study, therm1, therm2, therm3, therm4
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
ccap <- read_csv("data/raw/ccap2012.csv")
```

```
## New names:
## Rows: 43998 Columns: 3
## -- Column specification
## -------------------------------------------------------- Delimiter: "," dbl
## (3): ...1, caseid, gaytherm
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...1`
```

```r
nrow(gay_r) # number of observations
```

```
## [1] 11948
```

```r
nrow(ccap)
```

```
## [1] 43998
```

```r
names(gay_r) # names of columns
```

```
## [1] "study"     "treatment" "therm1"    "therm2"    "therm3"    "therm4"
```

```r
names(ccap)
```

```
## [1] "...1"     "caseid"   "gaytherm"
```

There are 11,948 observations in the gayreshaped dataset, and 43,998 observations in the ccap dataset. In the gayreshaped dataset, there are 6 columns: "treatment", "therm1", "therm2", therm3", "therm4". The therm variables measure thermometer scores from 0-100 toward gay men and lesbians. The ccap dataset has 3 columns: "..1", "caseid", and "gaytherm", where gaytherm measures thermometer attitudes toward gay men and lesbians.

Q2.2

```r
gay_baseline <- gay_r |>
  filter(study == 1)

gay_stats <- gay_baseline |>
  summarize(
    mean = mean(therm1, na.rm = TRUE),
    median = median(therm1, na.rm = TRUE),
    sd = sd(therm1, na.rm = TRUE)
  )

ccap_stats <- ccap |>
  summarize(
    mean = mean(gaytherm, na.rm = TRUE),
    median = median(gaytherm, na.rm = TRUE),
    sd = sd(gaytherm, na.rm = TRUE)
  )

comparison <- rbind( #combine both for side by side comparison
  cbind(dataset = "Gay Study", gay_stats),
  cbind(dataset = "CCAP", ccap_stats)
)

comparison
```

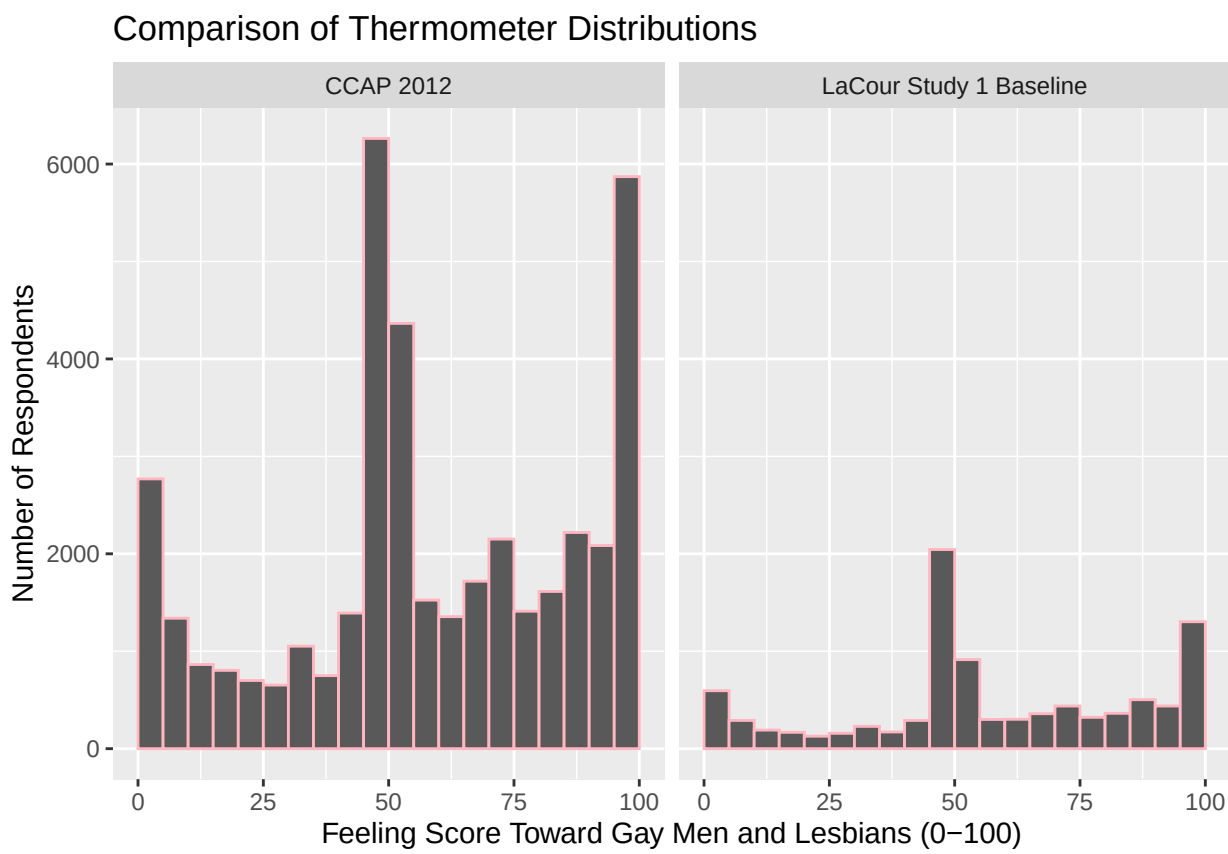
```
##     dataset     mean median       sd
## 1 Gay Study 58.37625     52 28.50288
## 2      CCAP 58.70517     54 29.36892
```

The scores in the Study 1 sample have a mean of 58.37625, median of 52, and standard deviation of 28.50288. In the 2012 CCAP dataset, the mean is 58.70517, median is 54, and standard deviation is 29.36892. These numbers are noticeably similar. Given the context of the study, this raises questions on whether the data has been fabricated to appear smoother.

Q2.3

```r
gay_histogram <- gay_r |>
  filter(study == 1) |>
  select(therm1) |>
  mutate(source = "LaCour Study 1 Baseline",
         therm = therm1) |>
  select(source, therm)

# CCAP thermometer (remove missing values)
ccap_histogram <- ccap |>
  filter(!is.na(gaytherm)) |>
  mutate(source = "CCAP 2012",
         therm = gaytherm) |>
  select(source, therm)
```

```r
combined_histogram <- rbind(gay_histogram, ccap_histogram) #combined into one dataset

ggplot(combined_histogram, aes(x = therm)) + # side by side historgram creation
  geom_histogram(breaks = seq(0, 100, by = 5), color = "lightpink") +
  facet_wrap(~ source) +
  labs(
    title = "Comparison of Thermometer Distributions",
    x = "Feeling Score Toward Gay Men and Lesbians (0-100)",
    y = "Number of Respondents"
  )
```



Shape-wise, both distributions look generally similar, with more concentration in respondents around the scores of 50 and 100 in both datasets. Because they appear so similar, this may be a red flag, as both these samples were supposed to have been collected independently from each other, so it is rather suspicious how they could appear so similar.

2.2 Re-test Reliability

Q2.4

```r
study1_control <- gay_r |>
  filter(study ==1, treatment == "No Contact") # for study 1 control group

study2_control <- gay_r |>
```

```r
  filter(study == 2, treatment == "No Contact") # for study 2

correlation_study1 <- cor(study1_control$therm1, study1_control$therm2, use = "complete.obs")
correlation_study2 <- cor(study2_control$therm1, study2_control$therm2, use = "complete.obs")

sd_study1 <- sd(study1_control$therm2 - study1_control$therm1, na.rm = TRUE)
sd_study2 <- sd(study2_control$therm2 - study2_control$therm1, na.rm = TRUE)

data.frame( # side by side view
  Study = c("Study 1", "Study 2"),
  Correlation = c(correlation_study1, correlation_study2),
  SD_Change = c(sd_study1, sd_study2)
)
```

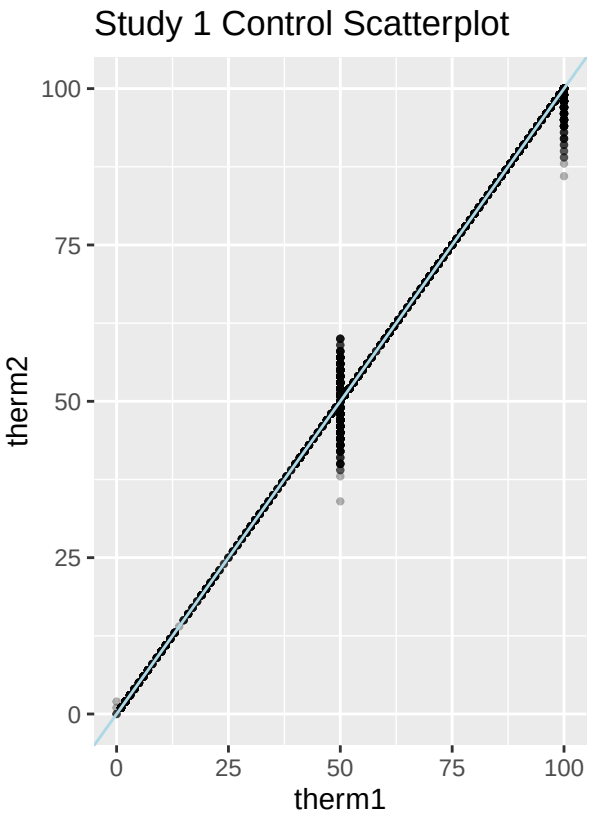
```
##     Study Correlation SD_Change
## 1 Study 1   0.9975817  1.987678
## 2 Study 2   0.9734449  6.605738
```

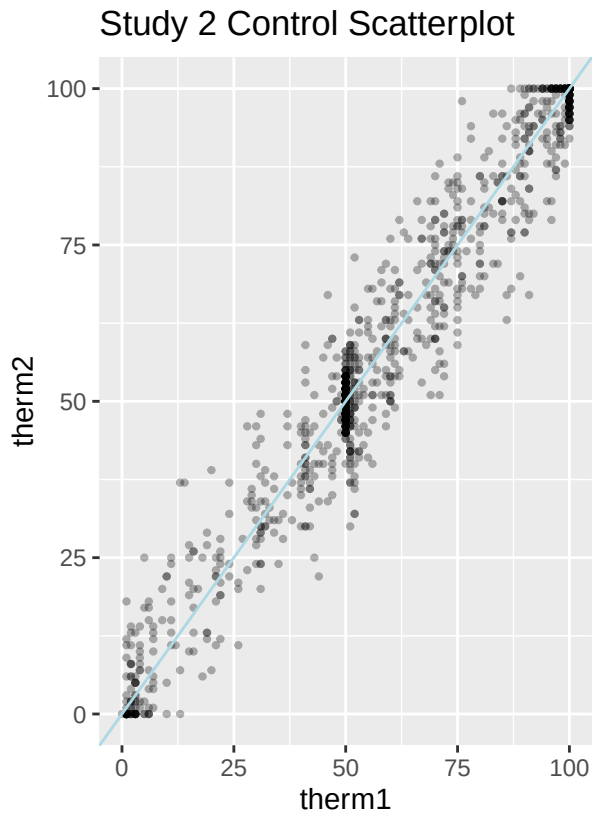
If we take a look at the results side by side, we can see that Study 2 shows a slightly more realistic pattern. It has a correlation of 0.973449, with a standard deviation of change of 6.605738. Meanwhile, Study 1's correlation is 0.9975817 when rounded, which is extremely high, with a standard deviation of change of 1.987678. What this suggests is that respondents' answers barely changed. Genuine survey data should yield more random variation, which Study 1 does not account for, so Study 1 appears comparatively more suspicious.

Q2.5

```r
library(patchwork) # for side by side layout
scatterplot1 <- ggplot(study1_control, aes(x = therm1, y = therm2)) + # create Study 1 scatterplot
  geom_point(alpha = 0.3, size = 0.8) +
  geom_abline(slope = 1, intercept = 0, color = "lightblue") +
  labs(title = "Study 1 Control Scatterplot",
       x = "therm1",
       y = "therm2")

scatterplot2 <- ggplot(study2_control, aes(x = therm1, y = therm2)) + # same steps for Study 2
  geom_point(alpha = 0.3, size = 0.8) +
  geom_abline(slope = 1, intercept = 0, color = "lightblue") +
  labs(title = "Study 2 Control Scatterplot",
       x = "therm1",
       y = "therm2")

scatterplot1 + scatterplot2
```

```
## Warning: Removed 568 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 164 rows containing missing values or values outside the scale range
## (`geom_point()`).
```





Upon comparing the side by side, it is alarming how near-perfectly aligned the data points fall on the reference line. Study 2 appears more realistic in that there is scatter around the line. Such results in Study 1 raises concerns on how the researchers managed to obtain such little variation amongst individial respondents.

2.3 Putting it Together

Q2.6

If we view the analyses altogether, it is evident that Study 1 data is most likely fabricated. First of all, it is rather alarming that the mean values of Study 1 versus the CCAP survey show very similar results (58.37625, 58.70517). This is unusual, as real data is collected independently. Next, when we retested Study 1 and Study 2, we found that Study 1 yielded a correlation of 0.9975817, while Study 2 had a correlation of 0.9734449. This tells us that the responses were essentially perfectly identical, which is simply cannot be plausible, as respondents' attitudes are bound to change over time. Lastly, when observing standard deviations, we found that the standard deviations of change for Study 1 versus Study 2 were 1.987678 and 6.605738 respectively. The small standard deviation in Study 1 indicates that there is very little variation in responses amongst respondents, which is unrealistic in genuine survey data.

3.1 Comparing Baselines

Q3.1

```r
study2 <- gay_r |> # filter for study 2
  filter(study == 2)

nrow(study2) # count total observations
```

```
## [1] 2441
```

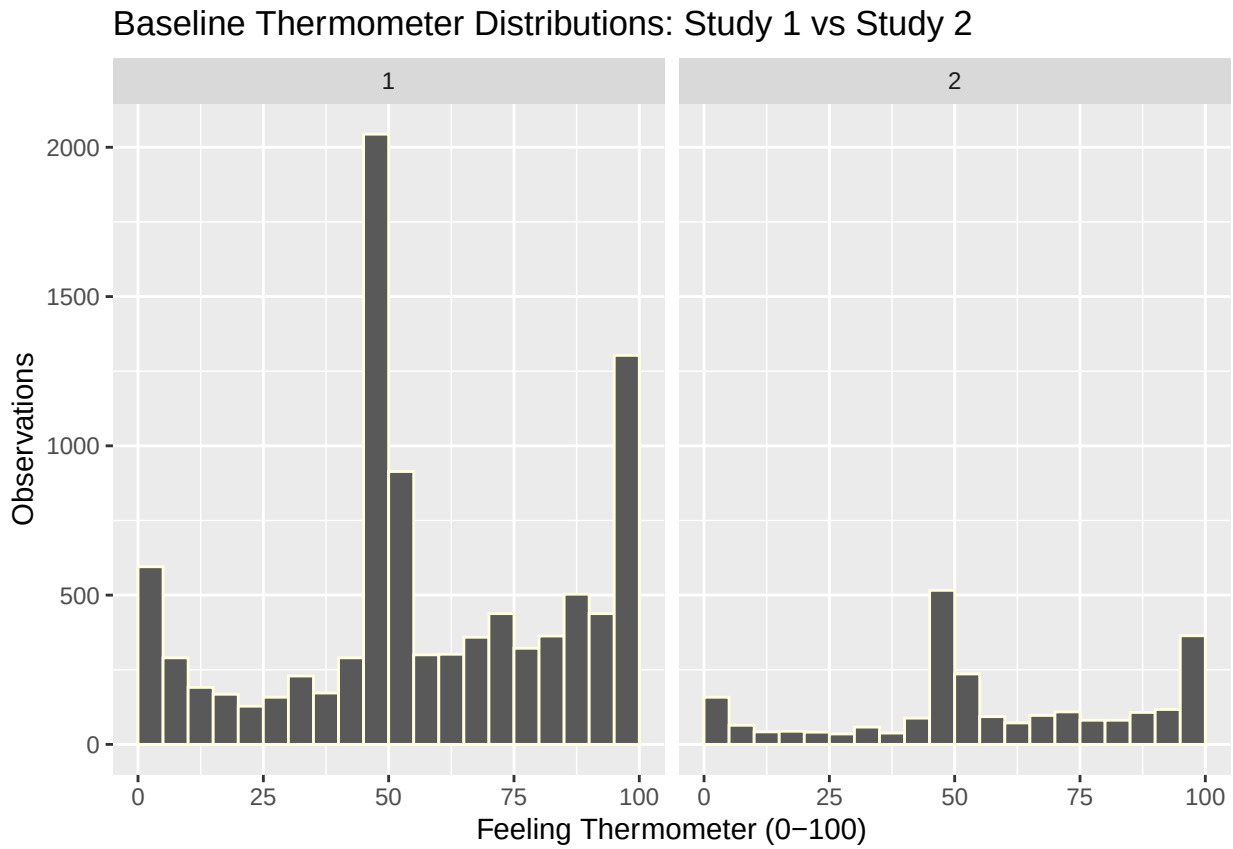
```r
study2 |>
  group_by(treatment) |>
  summarize(
    mean_therm1 = mean(therm1, na.rm = TRUE),
    sd_therm1 = sd(therm1, na.rm = TRUE),
    n = n()
  )
```

```
## # A tibble: 2 x 4
##   treatment                                  mean_therm1 sd_therm1     n
##   <chr>                                            <dbl>     <dbl> <int>
## 1 No Contact                                        57.9      28.7  1203
## 2 Same-Sex Marriage Script by Gay Canvasser         59.4      28.4  1238
```

We find that Study 2 has 2,441 observations. At the baseline, the mean thermometer score is 57.88612 in the No Contact group, and 59.35057 in the Same-Sex Marriage Script by Gay Canvasser group. The standard deviations are 28.69655 and 28.41578 for the two groups, respectively. There are 1203 observations in the No Contact group, and 1238 observations in the Gay Canvasser group.

Q3.2

```r
gay_r |>
  filter(study %in% c(1,2)) |> # keep observations for study 1 and 2
  ggplot(aes(x = therm1)) +
  geom_histogram(breaks = seq(0, 100, by = 5), color = "lightyellow") + # same as thermometer range
  facet_wrap(~ study) + # separate the panels
  labs(
    title = "Baseline Thermometer Distributions: Study 1 vs Study 2",
    x = "Feeling Thermometer (0-100)",
    y = "Observations"
  )
```



In terms of visual shape, Study 1 and Study 2 look very similar. They both peak in observations around the feeling thermometer score of 50 and 100. That being said, we must remember the fact that Study 1 has many more observations that Study 2, so it is odd how they can resemble each other even when the studies were conducted independently.

3.2 Checking Balance and Comparing Outcomes in Study 2

Q3.3

```
study2_check <- gay_r |> # now we do study 2
  filter(
    study == 2,
    treatment %in% c(
      "Same-Sex Marriage Script by Gay Canvasser",
      "No Contact"
    )
  )

study2_stats <- study2_check |> #mean and sd calculations
  group_by(treatment) |>
  summarize(
    mean_therm1 = mean(therm1, na.rm = TRUE),
    sd_therm1 = sd(therm1, na.rm = TRUE)
  )
```

```r
study2_stats
```

```
## # A tibble: 2 x 3
##   treatment                                   mean_therm1 sd_therm1
##   <chr>                                             <dbl>     <dbl>
## 1 No Contact                                         57.9      28.7
## 2 Same-Sex Marriage Script by Gay Canvasser          59.4      28.4
```

```r
mean_gayscript_study2 <- study2_stats$mean_therm1[
  study2_stats$treatment == "Same-Sex Marriage Script by Gay Canvasser"
]

mean_nocontact_study2 <- study2_stats$mean_therm1[
  study2_stats$treatment == "No Contact"
]

sd_gayscript_study2 <- study2_stats$sd_therm1[
  study2_stats$treatment == "Same-Sex Marriage Script by Gay Canvasser"
]

sd_nocontact_study2 <- study2_stats$sd_therm1[
  study2_stats$treatment == "No Contact"
]

SDMstudy2 <- (mean_gayscript_study2 - mean_nocontact_study2) /
  sqrt((sd_gayscript_study2^2 + sd_nocontact_study2^2) / 2)

SDMstudy2
```

```
## [1] 0.05128243
```

In study 2 at baseline, the mean and standard deviation of the No Contact group (control group) were 57.88612 and 28.69655. The Same-sex Marriage Script by Gay Canvasser group had a mean of 59.35057 and standard deviation of 28.41578. The standardized difference in means is 0.05128243. This is far below the 0.25 threshold, which indicates that the groups look well-balanced.

Q3.4

```r
study2_postconversation <- gay_r |>
  filter(
    study == 2,
    treatment %in% c(
      "Same-Sex Marriage Script by Gay Canvasser",
      "No Contact"
    )
  ) |>
  group_by(treatment) |>
  summarize(
    mean_therm2 = mean(therm2, na.rm = TRUE)
  )
```

```r
study2_postconversation
```

```
## # A tibble: 2 x 2
##   treatment                                   mean_therm2
##   <chr>                                             <dbl>
## 1 No Contact                                         57.6
## 2 Same-Sex Marriage Script by Gay Canvasser          61.2
```

```r
mean_gay_postconversation <- study2_postconversation$mean_therm2[ # difference
  study2_postconversation$treatment == "Same-Sex Marriage Script by Gay Canvasser"
]

mean_no_postconversation <- study2_postconversation$mean_therm2[
  study2_postconversation$treatment == "No Contact"
]

comparison_study2 <- mean_gay_postconversation - mean_no_postconversation

comparison_study2
```

```
## [1] 3.567885
```

In Study 2, the mean post-conversation thermometer score is 61.20311 for the Same-Sex Marriage Script by Gay Canvasser group and 57.63523 for the No Contact group. The difference in means is 3.567885 points. Since the groups were balanced at baseline, this difference suggests that the canvassing treatment did increase favorable attitudes toward gay men and lesbians.

3.3 Uncertainty Around the Difference

Q3.5

```r
study2_post_summary <- gay_r |> # filter to study 2
  filter(study == 2, treatment %in% c(
      "Same-Sex Marriage Script by Gay Canvasser",
      "No Contact")
  ) |>
  group_by(treatment) |> # calculations for values
  summarize(
    mean_therm2_study2 = mean(therm2, na.rm = TRUE),
    variance_therm2_study2 = var(therm2, na.rm = TRUE),
    n_therm2_study2 = sum(!is.na(therm2)) #sample size
  )

study2_post_summary
```

```
## # A tibble: 2 x 4
##   treatment            mean_therm2_study2 variance_therm2_study2 n_therm2_study2
##   <chr>                             <dbl>                  <dbl>           <int>
## 1 No Contact                         57.6                   817.            1039
## 2 Same-Sex Marriage S~               61.2                   827.            1093
```

```r
mean_gay_therm2_study2 <- study2_post_summary$mean_therm2_study2[ # means for each group
  study2_post_summary$treatment == "Same-Sex Marriage Script by Gay Canvasser"
]

mean_no_therm2_study2 <- study2_post_summary$mean_therm2_study2[
  study2_post_summary$treatment == "No Contact"
]

variance_gay_therm2_study2 <- study2_post_summary$variance_therm2_study2[ # variances for each group
  study2_post_summary$treatment == "Same-Sex Marriage Script by Gay Canvasser"
]

variance_no_therm2_study2 <- study2_post_summary$variance_therm2_study2[
  study2_post_summary$treatment == "No Contact"
]

n_gay_therm2_study2 <- study2_post_summary$n_therm2_study2[ #sample sizes for each group
  study2_post_summary$treatment == "Same-Sex Marriage Script by Gay Canvasser"
]

n_no_therm2_study2 <- study2_post_summary$n_therm2_study2[
  study2_post_summary$treatment == "No Contact"
]

# standard error calculations

SE_diff_therm2_study2 <- sqrt(
  (variance_gay_therm2_study2 / n_gay_therm2_study2) +
  (variance_no_therm2_study2 / n_no_therm2_study2)
)

SE_diff_therm2_study2
```

```
## [1] 1.242138
```

```r
diff_means_therm2_study2 <-
  mean_gay_therm2_study2 - mean_no_therm2_study2 # must find difference in means for CI calculation

lower <- diff_means_therm2_study2 - 1.96 * SE_diff_therm2_study2 # confidence interval calculations

upper <- diff_means_therm2_study2 + 1.96 * SE_diff_therm2_study2

c(lower, upper)
```

```
## [1] 1.133294 6.002475
```

The standard error is 1.242138, and the 95% confidence interval is 1.133294, 6.002475. This interval does NOT include 0. Thus, we can conclude that the difference between Gay Canvasser and No Contact groups is statistically significant, at the 5% level. We can say that the act of canvassing increased feeling scores by approximately 1-6 points in study 2.

3.4 Tracking Attitude Over Time

3.6

```
study2_control <- gay_r |> # filter for study 2 only
  filter(
    study == 2,
    treatment == "No Contact"
  )

correlation12 <- cor(study2_control$therm1, # correlation between wave1 and wave2
               study2_control$therm2,
               use = "complete.obs")

correlation13 <- cor(study2_control$therm1, # correlation between wave1 and wave3
               study2_control$therm3,
               use = "complete.obs")

correlation14 <- cor(study2_control$therm1, # correlation between wave1 and wave4
               study2_control$therm4,
               use = "complete.obs")

correlation12
```

```
## [1] 0.9734449
```

```
correlation13
```

```
## [1] 0.9594085
```

```
correlation14
```

```
## [1] 0.9709017
```

The correlation between therm1 and therm2 is 0.9734449, between therm1 and therm3 is 0.594085, and between therm1 and therm4 is 0.9709017. These correlations are extremely high and do not show a clear pattern of decay over time, as the correlation with therm4 is nearly the same as with therm2 In real survey data, we would typically expect correlations to decline as waves become more distant due to natural attitude change, so this stable pattern does raise concerns about the authenticity of the data.

Q3.7

Study 1 and 2 differ and size and how realistic the results are. In study 2, the post-conversation treatment effect is 3.567885 on the 0-100 thermometer scale, with a 95% confidence interval of 1.133294, 6.002475. This means that there is a statistically significant but uncertain effect. In contrast, study 1 showed only a 0.10 difference on the 1-5 scale at wave 2, despite the data appearing suspiciously smooth. Usually, real survey data would contain potential variation and errors, so treatment effects should be rather modest. It is important that real treatment effects are smaller and messier because peoples' attitudes are constantly changing. It's actually quite cool to see how we can pick out fabrications in data through analyses tools as simple as histograms and mean calculations. Sometimes, it takes visualization or direct side by side comparison of data to debunk patterns that look too "perfect" to be genuine.

4 Part 4: Formalizing the Evidence with Regression

4.1 Regression as a Fraud Detector

```r
fit_s1 <- lm(therm2 ~ therm1, data = study1_control) # fit the regression

summary(fit_s1) # display regression
```

```
##
## Call:
## lm(formula = therm2 ~ therm1, data = study1_control)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.8631  -0.1121   0.1746   0.4312  10.1369
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.240430   0.066000   3.643 0.000273 ***
## therm1      0.992454   0.001012 980.624  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.976 on 4668 degrees of freedom
##   (568 observations deleted due to missingness)
## Multiple R-squared:  0.9952, Adjusted R-squared:  0.9952
## F-statistic: 9.616e+05 on 1 and 4668 DF,  p-value: < 2.2e-16
```

The reported slope is 0.992454, the R-squared is 0.9952, and residual standard error is 1.976.

Q4.2

```r
study2_control <- gay_r |> # same process but filter for study 2
  filter(study == 2, treatment == "No Contact")

fit_s2 <- lm(therm2 ~ therm1, data = study2_control)

summary(fit_s2)
```

```
##
## Call:
## lm(formula = therm2 ~ therm1, data = study2_control)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -22.7102  -3.4793   0.0363   3.5092  22.8455
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) 1.561897   0.457098   3.417 0.000658 ***
## therm1      0.968662   0.007074 136.935  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.547 on 1037 degrees of freedom
##   (164 observations deleted due to missingness)
## Multiple R-squared:  0.9476, Adjusted R-squared:  0.9475
## F-statistic: 1.875e+04 on 1 and 1037 DF,  p-value: < 2.2e-16
```

The Study 1 model produces a slope of 0.992454 and an R-squared of 0.9952, indicating that therm1 almost perfectly predicts therm2. In contrast, the Study 2 model has a lower slope of 0.968662 and a lower R-squared of 0.9476, along with a larger residual standard error. This pattern is consistent with what we would expect in real data, where responses change somewhat over time. The nearly perfect relationship in Study 1 is thus suspicious and consistent with the pattern we would expect if the follow-up values were artificially generated.

4.2 Comparing Groups With Regression

Q4.3

```r
study2_groups <- gay_r |> # filter Study 2 to only gay canvasser
  filter(
    study == 2,
    treatment %in% c("Same-Sex Marriage Script by Gay Canvasser", "No Contact")
  )

study2_groups <- study2_groups |> # create binary treatment variable
  mutate(
    treated = ifelse(treatment == "Same-Sex Marriage Script by Gay Canvasser", 1, 0)
  )

fit_groups <- lm(therm2 ~ treated, data = study2_groups) # comparing treated vs control

summary(fit_groups)
```

```
##
## Call:
## lm(formula = therm2 ~ treated, data = study2_groups)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -61.203 -15.635  -3.203  24.365  42.365
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  57.6352     0.8895  64.794  < 2e-16 ***
## treated       3.5679     1.2423   2.872  0.00412 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 28.67 on 2130 degrees of freedom
```

```
##   (309 observations deleted due to missingness)
## Multiple R-squared:  0.003857,   Adjusted R-squared:  0.00339
## F-statistic: 8.248 on 1 and 2130 DF,  p-value: 0.00412
```

The estimated coefficient on treated (beta 1 hat) is 3.5679, with a standard error of 1.2423, a t-statistic of 2.872, and a p-value of 0.00412. This value is essentially identical to the difference in means we calculated earlier of 3.567885 points, which demonstrates that a regression with a binary treatment variable yields the same result as comparing group means directly. The intercept of 57.6352 represents the average therm2 score for the control group. The coefficient on treated therefore captures the difference between the treatment and control groups. Since the p<0.05, which is the significance level, we are able to conclude that the treatment effect is statistically significant.

4.3 Controlling for Baseline Attitudes

Q4.4

```r
study2_groups <- gay_r |>
  filter(
    study == 2,
    treatment %in% c("Same-Sex Marriage Script by Gay Canvasser", "No Contact")
  ) |>
  mutate(
    treated = ifelse(treatment == "Same-Sex Marriage Script by Gay Canvasser", 1, 0)
  )

fit_groups_control <- lm(therm2 ~ treated + therm1, data = study2_groups) #adding therm 1 as a control

summary(fit_groups_control)
```

```
##
## Call:
## lm(formula = therm2 ~ treated + therm1, data = study2_groups)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -27.194  -4.476  -0.476   3.688  35.558
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1.727135   0.423275    4.08 4.66e-05 ***
## treated     2.458857   0.341052    7.21 7.76e-13 ***
## therm1      0.965808   0.005973  161.69  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.87 on 2129 degrees of freedom
##   (309 observations deleted due to missingness)
## Multiple R-squared:  0.925,  Adjusted R-squared:  0.9249
## F-statistic: 1.313e+04 on 2 and 2129 DF,  p-value: < 2.2e-16
```

After adding the baseline attitude therm1 as a control, the estimated treatment effect fell from 3.5679 to 2.458857 points. Additionally, the standard error dropped from 1.2423 to 0.341052, which is a 72.6 percent

reduction. Despite the smaller coefficient, the treatment effect remains statistically significant, with a p-value<0.001. Including the baseline attitudes improves precision because therm1 strongly predicts therm2. So, we can observe that accounting for respondents' initial attitudes explains a large portion of the variation in follow-up responses, which reduces residual noise and produces a much smaller standard error.

Q4.5

```
personA <- data.frame(treated = 1, therm1 = 50) # create the hypothetical people
personB <- data.frame(treated = 0, therm1 = 50)

predicted_a <- predict(fit_groups_control, newdata = personA)
predicted_b <- predict(fit_groups_control, newdata = personB)

predicted_a
```

```
##        1
## 52.47638
```

```
predicted_b
```

```
##        1
## 50.01752
```

```
predicted_a - predicted_b # difference in wave 2 attitudes
```

```
##        1
## 2.458857
```

The R-squared increases greatly from (0.003857 to 0.925 after adding therm1 to the model. What this shows is that baseline attitudes account for most of the variation in follow-up attitudes, while the treatment explains only a small portion.

The predicted wave 2 attitude for person A is 52.47638, while the predicted value for person B is 50.01752. The predicted difference between them is 2.45886 points. Since both respondents start with the same baseline attitude where therm1 = 50, this difference reflects the estimated treatment effect from the regression model. Essentially, the model predicts that receiving the treatment increases the follow-up thermometer score by 2.45886 points compared to the control group.